

Treat the following exercises:

I-(4 pts)

Design the UML meta-model with a class diagram.

For example: a UML model can contain classes, interfaces and relationships. A class can be abstract or not. A class is described by attributes and methods.

An attribute has a type that can be (simple, class or interface). An attribute also has modifiers (public, private, protected, static, final, etc.). A method has a signature formed of modifiers, optionally a return type, a method name and optionally one or more parameters. A method may be abstract or not.

A relationship is an association (binary, ternary or unary), an aggregation, a composition or an inheritance from one or more classes.

N.B: Add all your knowledge in UML to the class diagram.

II- (6 pts)

The drawing editor is an interactive graphics editor. With it, users can create and edit drawings composed of lines, rectangles, ellipses and text.

Tools control the mode of operation of the editor. One tool is active at any time. Two kinds of tools exist: the selection tool and the creation tools. When the selection tool is active, existing drawing elements can be selected with the cursor. One or more drawing elements can be selected and manipulated. If several drawing elements are selected, they can be manipulated as if they were a single element. Elements that have been selected in this way are referred to as the current selection. The current selection is indicated visually by displaying the control points for the element. Clicking on and dragging a control point modifies the element with which the control point is associated.

When a creation tool is active, the current selection is empty. The cursor changes in different ways according to the specific creation tool, and the user can create an element of the selected kind.

The text creation tool: the position of the first character of the text is determined by where the user clicks the mouse button. The creation tool is no longer active when the user clicks the mouse button outside the text element. The control points for a text element are at the four corners of the region within which the text is formatted. Dragging the control points changes this region.

The other creation tools allow the creation of lines, rectangles and ellipses. The appropriate element starts to be created when the mouse button is pressed, and is completed when the mouse button is released. These two events create the start point and the stop point.

The line creation tool creates a line from the start point to the stop point. These are the control points of the line. Dragging a control point changes the associated end point of the line.

The rectangle creation tool creates a rectangle such that these points are diagonally opposite corners. These points and the other corners are the control points. Dragging a control point changes the associated corner, and can change other corners as well.

The ellipse creation tool creates an ellipse fitting within the rectangle defined by the two points described above. The major radius is one half the width of the rectangle, and the minor radius is one half the height of the rectangle. The control points are at the corners of the bounding rectangle. Dragging a control point changes the associated corner.

Draw a class diagram of the drawing editor.

III- (2 pts)

The ESU university wants to automate its registration system.

- The head of registration office establishes the courses curriculum for a semester
 - A course may be offered several times
- The students must select 4 primary courses and 2 secondary courses
- Once a student has registered for a semester, the billing system is notified
- The students can use the system to change their choice for a certain period of time after their registration
- The teachers use the system to view their schedule
- The users of the registration system receive passwords that are necessary to the identification procedure

Draw the use case diagram of the registration system.

IV- (3 pts)

Draw a state diagram of a simple calculator. The interface of the calculator is composed of 10 buttons with digits, and 4 buttons with the basic operations (+, -, *, /). The button "C" resets the display. The button "=" displays the result. And there are two additional buttons "ON" and "OFF":

V- (3 pts)

An access control device by magnetic card to a photocopier machine is equipped with a display screen that can display the following messages:

- "INSERT YOUR CARD": when the device is idle.
- "WAIT": while the device reads the code of a card inserted.
- "INVALID CARD": when the code is not recognized, then the card is automatically ejected.
- "ENTER YOUR CODE": when the card has been read.
- "CODE FAIL": If the code entered is not identical to the code read; the card is automatically ejected.
- "USE IN PROGRESS": when the code entered is correct.

The user may at any time activate a button which causes the ejection of the card. After any eject card, the device displays "INSERT CARD".

Propose the state diagram of the card reader.

VI- (2 pts)

When customer requests a room, the hotel employee looks for room availability, then he inserts customer's data and verifies if the provided credit card number has enough credit to pay the room: in this case the booking is confirmed and we wait customer arrival. The Hotel information system allows customer to modify or cancel his/her reservation. If customer arrives he uses the room and then he pays, in case of no-attendance, a penalty is accounted on his credit card and the booking is canceled.

Draw the activity diagram.
